

Original Article

DOI 10.1007/s12206-026-0243-8

Keywords:

- Graph algorithm
- Impeller
- Normal of point clouds
- Principal components analysis
- Weld seam recognition

Correspondence to:

Dongmin Li
ldm753@163.com

Citation:

Li, D., Wang, Y., Wang, Z. (2026). Automatic recognition on impeller shape and weld seam based on normal of point clouds and PCA. *Journal of Mechanical Science and Technology* 40 (3) (2026) 2039–2046.
<https://doi.org/10.1007/s12206-026-0243-8>

Received January 23rd, 2025

Revised September 12th, 2025

Accepted October 28th, 2025

† Recommended by Editor
Hae-Jin Choi

Automatic recognition on impeller shape and weld seam based on normal of point clouds and PCA

Dongmin Li^{1,2}, Yu Wang¹ and Zhengyong Wang¹

¹College of Intelligent Equipment, Shandong University of Science and Technology, Tai'an 271019, China,
²Research and Development Section, Shandong Judoshi Energy Technology Co., LTD., Tai'an 271000, China

Abstract

The accuracy of the recognition on weld seam between impeller hub and fan blade are still low due to the texture characteristic of welded workpieces and lighting, thus the recognition on weld seam based on point clouds needs to be improved. Besides, the existing recognition methods are generally performed based on the point clouds of weld seam of flat surface, which are not suitable for the curved fan blade, therefore, we propose an automatic recognition method on the weld seam to improve the accuracy. Firstly, the original point clouds are denoised and compressed by Gaussian filter and voxel filter, respectively, and the point cloud model is divided into two types using region growth method. Next, the shape of the welding surface was recognized by principal components analysis (PCA) and normal distribution theory, and the position of the weld seam was obtained by least square method, furthermore, the welding path was determined by graph algorithm and bubble sorting method. The welding experimental results showed that the weld seam between the fan blade with different shapes and impeller hub was recognized successfully, and the offset error between the weld seam position obtained using our algorithm and the actual position was less than 1 mm, which improved accuracy of the recognition on weld seam. Besides, the weld seam meets the requirements of the welding technology based on the weld seam inspection.

1. Introduction

Welding technology is applied to connect two metal plates together generally, and the accuracy of recognition on weld seam plays a crucial role in the welding quality and product performance [1, 2]. The existing recognition on weld seam is performed as follows: the laser line projects on the welding position to get a two-dimensional picture of mutation points to extract the weld seam. Wang et al. designed a weld seam image processing system based on laser vision, and the coordinate of the center of the weld seam was obtained using the combination of the open operation and the morphology operation [3]. Guo and Zhang designed a mobile robot based on laser vision sensor to identify the weld seam in the narrow space, and the weld seam was determined according to the inflection point of the laser line [4, 5]. Li et al. proposed a method of tracking and recognition on weld seam automatically based on the structured light. The grayscale distribution of laser fringes was tracked using Kalman filter, furthermore the weld contour within a small range was searched to decrease disturbance [6]. The above methods have improved the efficiency of recognition on weld seam to a certain extent. However, the stability and accuracy of the recognition on weld seam are still low due to the effect of the texture characteristic of the welded workpiece and lighting [7, 8]. Therefore, the recognition on weld seam based on point clouds needs to be improved to meet the actual requirements.

Point cloud is defined as a set composed of many points on the surface generated by collection of 3D (three dimensional) scanner or depth camera, which can capture the geometric shape of objective surface meanwhile is independent of texture of the object. Takubo et al. used a

binocular depth camera to identify weld seam at the optimal position [9]. The two welding surfaces were captured by in both horizontal and vertical directions, respectively. Furthermore, the weld seam was obtained according to the intersection line of two surfaces. However, this method is only suitable for the flat welded workpiece since it cannot capture the curved workpiece in a fixed diagonal posture. Wang et al. proposed a visual guidance method for welding robots based on object recognition and pose estimation [10]. This method recognized the welded workpiece by matching the welding point cloud in the welding model library. Furthermore, it generated the welding trajectory and the pose of the welded workpiece. Geng et al. proposed a rough extraction algorithm of point clouds for weld seams of impeller blades based on distance [11]. The farthest point was obtained by comparing the farthest points away from the 3D camera in the same segment, thus the rough extraction of the weld seam point clouds was accomplished. Cao et al. recognized the weld seam based on 2D (two-dimensional) grayscale data and 3D point cloud [12].

Unlike the recognition methods on 2D image of weld seam, the key to identify the point clouds of weld seam is to fit the point clouds of the welding surface according to the shape characteristic of the workpiece. Yang et al. designed a vision sensor based on digital light processing (DLP), and the weld seam position was determined by the fit of the point clouds and welding surface, furthermore the welding path and welding pose were determined [13]. Liu et al. proposed a method of recognition on welding path based on point cloud reconstruction, this method divided the point clouds into two welding surfaces, and obtained the weld seam of the workpiece using boundary extraction method [14]. Jia et al. obtained the pose of the weld seam using online rapid extraction method on the pose for 3D polyline weld seam to identify the weld seam [15]. Zhang et al. proposed an extracting method for the flat weld seam based on 3D vision [16].

In summary, the existing recognition method on weld seam is generally performed based on the point clouds of weld seam of flat surface, which is not suitable for the surface such as impeller fan blade, thus we propose an automatic recognition method on the weld seam between the fan blade with different shapes and impeller hub. Firstly, the point clouds of impeller are divided into point cloud of fan blade and point cloud of impeller hub. Furthermore, the shape type of the welding surface is recognized. Finally, the initial position and path of the weld seam is determined and sent to the robotic controller.

2. Pretreatment of welding surface point cloud

The point cloud of the workpiece generated by camera scanning is inclined to generate noise point due to various disturbances, which damages the point cloud processing. Therefore, the point clouds need to be preprocessed. Besides, the original dense point clouds lead to high cost in calculation memory and slow processing speed, thus the downsampling operation is performed to decrease the number of the point clouds and improve

the recognition efficiency.

2.1 Noises removal of point clouds

Because the density of the outlier noise points generated by camera scanning the workpiece is much low than that of the point clouds of the workpiece surface, the noise can be removed by Gaussian filter. Gaussian filter means to take a weighted average operation on the adjacent points near the target point in the workpiece point clouds using Gaussian function, and smooth point clouds of the workpiece [17]. The point clouds and outlier noise in the workpiece surface are recognized since lots of characteristics in the workpiece surface are included in the dense point cloud area.

2.2 Point cloud filtering

To obtain high-precision weld seam point clouds, some methods such as voxel centroid filtering, voxel center filtering, were adopted. Comparatively, voxel centroid filtering method can extract multiple scale and multiple level information of local features and has higher accuracy. Therefore, voxel centroid filtering method is adopted to decrease sampling rate and the amount of the point clouds [18].

2.3 Segmentation of welding surface

To improve the extraction effect of 3D point clouds of weld seam of impeller, region growing method is used to divide the initial point clouds into impeller hub point clouds and fan blade point clouds according to the intersection line between two intersection welding surfaces [19].

3. Recognition on weld seam

The welding of impeller used for ventilator is an operation to weld the impeller hub and fan blade, where, the shape of the fan blade is a flat or curved surface. Based on the point clouds fitting for the different shapes of fan blades, a fitting algorithm of point clouds is proposed based on point clouds normal of weld seam, as shown in Fig. 1.

Firstly, the normal direction of the point clouds is calculated by principal components analysis (PCA) [20]. Next, the statistics on the normal distribution is performed according to all the normal direction of the point clouds. When the distribution probability of the normal angle within the standard deviation is over the threshold, the point clouds is regarded as a flat surface point cloud, otherwise, a curved surface point cloud. Next, according to the recognition results, the corresponding fitting equation is used to fit the point clouds, and the ideal welding path is obtained based on the intersection line between the two surfaces. Furthermore, the distance between every point and the ideal welding path is calculated, when it is within the threshold, this point is regarded as an actual weld seam position. Finally, bubble sorting method is used to sort the welding points, and the results are output to

the controller.

3.1 Determination of normal of point clouds

For the different shape of fan blade, its local surface P can be fitted by least square method, and the normal expression of the point clouds in this surface is obtained [21]. For an arbitrary point p_i in the point clouds, assuming there are k points within its adjacent domain, when the normal vector of the local surface $ax+by+cz+d=0$ becomes the vector $n = (a, b, c)$, the above relationship can be represented by Eq. (1).

$$\begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ x_k & y_k & z_k & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (1)$$

Furthermore, the local surface P can be represented by Eq. (2).

$$P(n, d) = \arg \min_{(n, d)} \sum_{i=1}^k (n \cdot p_i - d)^2 \quad (2)$$

Besides, a curved surface can be regarded as the combination of m local surfaces, and the normal direction is the direction with the minimum principal component, i.e., the eigenvector corresponding to the minimum eigenvalue obtained by PCA, therefore, the normal of the curved surface can be solved by PCA.

According to PCA, the covariance matrix of the local surface is calculated by Eq. (3).

$$M_p = \frac{1}{k} \sum_{i=1}^k (p_i - \bar{p})(p_i - \bar{p})^T \quad (3)$$

Where, $\bar{p}$ is the mean value of the domain point clouds of the point p_i , as shown in Eq. (4).

$$\bar{p} = \frac{1}{k} \sum_{i=1}^k p_i = \left(\frac{1}{k} \sum_{i=1}^k x_i, \frac{1}{k} \sum_{i=1}^k y_i, \frac{1}{k} \sum_{i=1}^k z_i \right) \quad (4)$$

The eigenvector corresponding to the minimum eigenvalue obtained by Eq. (3) is regarded as the normal vector of the local surface. The normal vector set $N = \{n_i | i = 1, 2, 3, \dots, m\}$ of the curved surface is composed of the normal vector of m local surfaces. The covariance matrix M_p shown in Eq. (3) is fitted and becomes the normal distribution model [22], as shown in Eq. (5). Among that, μ denotes the mean value of the normal vector, as shown in Eq. (6).

$$f(N) = \frac{1}{\sqrt{(2\pi) \det(M_p)}} \exp \left(-\frac{1}{2} (N - \mu)^T M_p^{-1} (N - \mu) \right) \quad (5)$$

$$\mu = \frac{1}{m} \sum_{i=1}^m n_i \quad (6)$$

As we all know, the normal direction for all points in the surface is same. When the distribution probability of the normal vectors within the standard deviation range is over 90 %, i.e., $\mu \pm \sigma > 90\%$, this point cloud belongs to flat surface, otherwise, curved surface. Among that, σ denotes standard deviation, i.e., error from the basic deviation angle of the normal, and the threshold is determined by the minimum proportion of non-edge point clouds to the total point clouds.

3.2 Fitting of welding surface

Fitting of welding surface is a process to construct a continuous

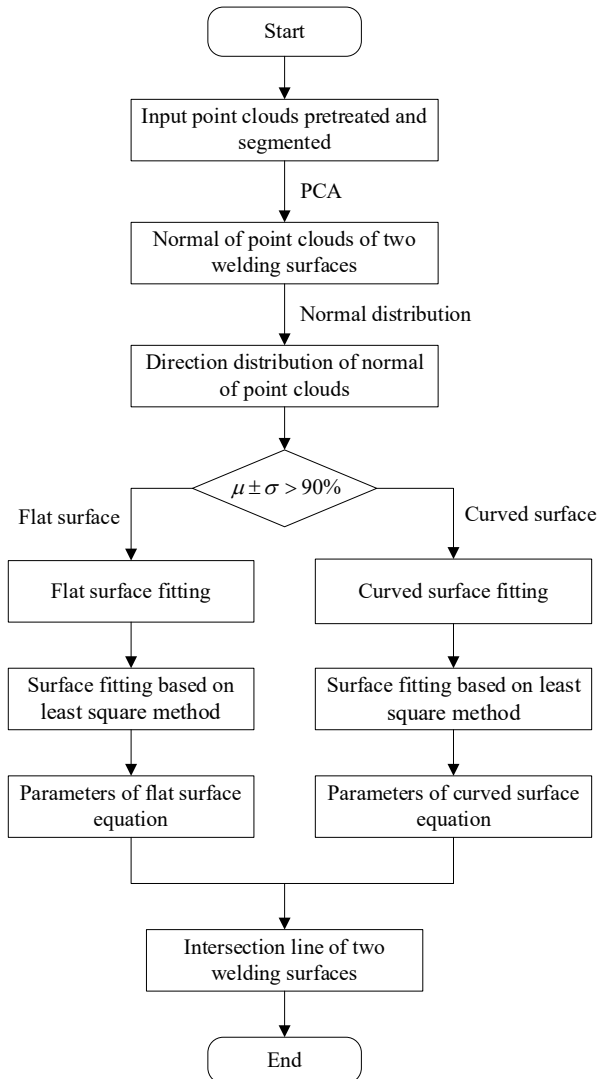


Fig. 1. Recognition process of weld seam.

surface model from the discrete point cloud set, and obtain the optimal flat or curved surface fitted with the point clouds shape. Based on the characteristic of the welding surface, the point clouds of the flat surface $\{(x_i, y_i, z_i) | i = 1, 2, 3, \dots, N\}$ is fitted by least square method [23], and the main steps are as follows.

(1) Loss function of the welding surface

The distance D_i between the point from the point clouds of the flat surface and the flat surface $ax+by+cz+d=0$ is defined by Eq. (7). Among that, a , b , c , and d denote the equation parameters of the flat surface, and the optimal fitting parameters are obtained based on the minimum distance got from Eq. (7).

$$D_i = \frac{|ax_i + by_i + cz_i + d|}{\sqrt{a^2 + b^2 + c^2}} \quad (7)$$

The loss function of welding surface is shown in Eq. (8), and the minimum distance is obtained by the loss function calculation.

$$L(a, b, c, d) = \sum_{i=1}^N (ax_i + by_i + cz_i + d)^2 \quad (8)$$

(2) Equation parameters of the flat surface

According to the relationship between the loss function and the equation parameters of the flat surface, the partial derivative of the loss function to the equation parameters of the flat surface is calculated using Eq. (9).

$$\begin{cases} \frac{\partial L}{\partial a} = -2 \sum_{i=1}^N (ax_i + by_i + cz_i + d) x_i \\ \frac{\partial L}{\partial b} = -2 \sum_{i=1}^N (ax_i + by_i + cz_i + d) y_i \\ \frac{\partial L}{\partial c} = -2 \sum_{i=1}^N (ax_i + by_i + cz_i + d) z_i \\ \frac{\partial L}{\partial d} = -2 \sum_{i=1}^N (ax_i + by_i + cz_i + d) \end{cases} \quad (9)$$

To minimize the loss function, all the partial derivatives of the loss function to the equation parameters of the flat surface are set to 0, as shown in Eq. (10).

$$\begin{cases} \sum_{i=1}^N (ax_i + by_i + cz_i + d) x_i = 0 \\ \sum_{i=1}^N (ax_i + by_i + cz_i + d) y_i = 0 \\ \sum_{i=1}^N (ax_i + by_i + cz_i + d) z_i = 0 \\ \sum_{i=1}^N (ax_i + by_i + cz_i + d) = 0 \end{cases} \quad (10)$$

Next, Eq. (10) is simplified, as is shown in Eq. (11). Furthermore, the optimal equation parameters a , b , c , and d of the flat surface can be obtained by Eq. (11).

$$\begin{cases} a \sum_{i=1}^N x_i^2 + b \sum_{i=1}^N x_i y_i + c \sum_{i=1}^N x_i z_i + d \sum_{i=1}^N x_i = 0 \\ a \sum_{i=1}^N x_i y_i + b \sum_{i=1}^N y_i^2 + c \sum_{i=1}^N y_i z_i + d \sum_{i=1}^N y_i = 0 \\ a \sum_{i=1}^N x_i z_i + b \sum_{i=1}^N y_i z_i + c \sum_{i=1}^N z_i^2 + d \sum_{i=1}^N z_i = 0 \\ a \sum_{i=1}^N x_i + b \sum_{i=1}^N y_i + c \sum_{i=1}^N z_i + d \sum_{i=1}^N 1 = 0 \end{cases} \quad (11)$$

(3) Equation parameters of the curved surface

When the welding surface is a curved surface, the surface equation needs to be replaced with $x^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$, and its loss function is shown in Eq. (12).

$$\begin{aligned} L(A, B, C, D, E, F, G, H, I, J) \\ = \sum_{i=1}^N (Ax_i^2 + By_i^2 + Cz_i^2 + Dx_i y_i + Ex_i z_i + K)^2 \end{aligned} \quad (12)$$

Among that, $K = Fy_i z_i + Gx_i + Hy_i + Iz_i + J$.

Similarly, the equation parameters of the curved surface A , B , C , D , E , F , G , H , I , and J are obtained.

The fitting result of the curved welding surface is shown in Fig. 2.

3.3 Extraction of weld seam

The weld seam can be obtained based on the intersection line between two fitted welding surfaces, which is the theoretical weld seam by image processing. Besides, the weld seam generated from the image point clouds deviates from the theoretical weld seam due to uneven workpiece surface. To eliminate the above deviation, a series of point clouds within the threshold are obtained based on the distance between the point clouds of the

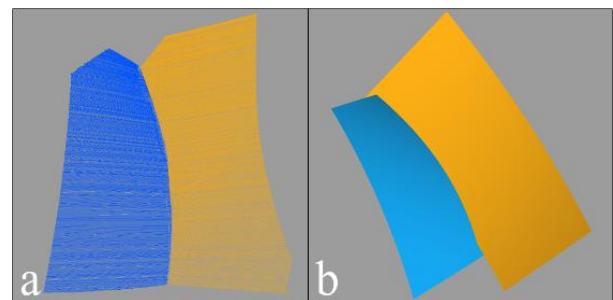


Fig. 2. Fitted curved welding surface: (a) point clouds of curved surface; (b) fitted curved surface.

impeller and the intersection line, thus they construct the optimal weld seam.

Besides, the points of the weld seam need to be sorted and constitute the welding path. Firstly, the initial point of the welding path is determined based on graph algorithm [24], here, the so-called graph is a data structure indicating the “connection” relationship between each pair of elements. The point clouds out of order can be regarded as the boundary points in the graph. A weighted undirected graph can be established by connecting two arbitrary and non-intersecting points. Among that, w denotes the weight and the distance between two points, too.

Furthermore, a random point is selected as the source point, and the distance between this point and the adjacent points is calculated, thus the adjacent point with the minimum distance is taken as a new source point, meanwhile the old source point is removed. All the points are traversed using the above method, and the last point reserved is the initial point of the weld seam. Therefore, the shortest path is represented in Eq. (13).

$$d(v) = \min(d(v), d(u) + w(u, v)) \quad (13)$$

Among that, $d(v)$ denotes the distance between the source point and the adjacent points, $d(u)$ denotes the distance between the old source point u and the new source point v , and $w(u, v)$ denotes the weight of the side between two points.

Finally, the distances between all weld seam points and the initial point are calculated, and sorted in order from small to large by bubble sort method [25] to obtain the welding path.

4. Experiments and analysis

4.1 Establishment of experimental platform

The welding experimental platform is built as Fig. 3, and its hardware consists of FANUC M-10id/12 six-axis robot, TRM welding gun, and full-line structured light 3D camera. Its software mainly consists of Windows 10 operating system, PCL point clouds library, and Visual Studio 2022. Besides, the welded impeller is made of carbon steel.

4.2 Recognition on shape of welding surface

To verify the effectiveness of recognition on shape of welding surface with our algorithm, the welded workpieces with flat surface shape and curved surface shape are scanned respectively



Fig. 3. Welding experimental platform for impeller.

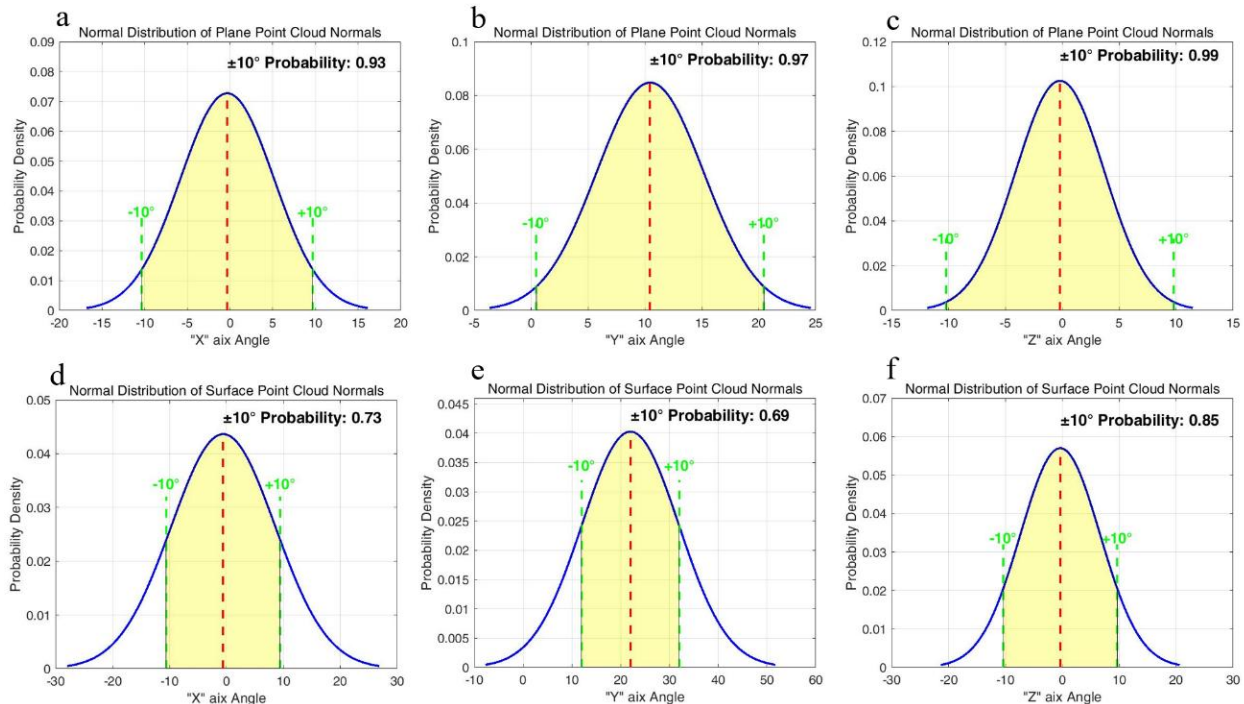


Fig. 4. Normal angle distribution of point clouds: Normal of point clouds of flat surface based on (a) x-axis; (b) y-axis; (c) z-axis; normal of point clouds of curved surface based on (d) x-axis; (e) y-axis; (f) z-axis.

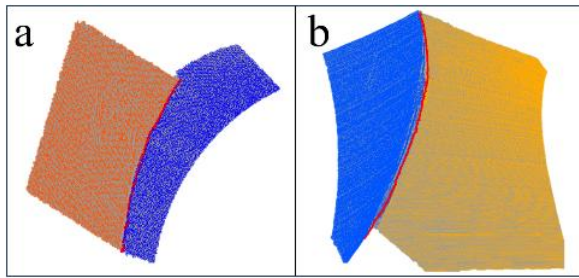


Fig. 5. Point clouds of weld seam: (a) flat fan blade; (b) curved fan blade.

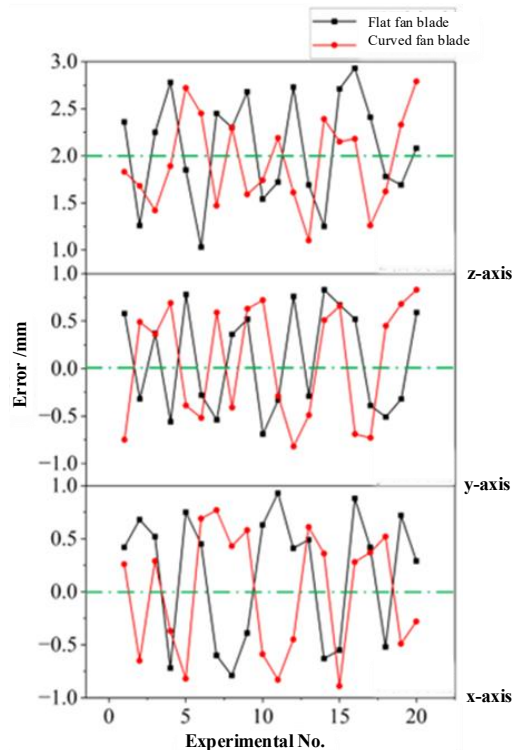


Fig. 6. Position errors of weld seam.

by the welding robot according to Sec. 2.2 “fitting of the welding surface”, and the normal distribution of point clouds of two shapes of welding surfaces is done by normal distribution theory, as shown in Fig. 4. Among that, the value in the horizontal axis represents the deviation angles of normal in direction of x , y , and z , and the value in the vertical axis represents the probability density of the deviation angles. Besides, the red dashed line represents the mean μ of the normal angle, and the green dashed line represents the standard deviation σ .

According to Figs. 4(a), (b), and (c), the normal angle of the point clouds is higher based on x -axis, and lesser based on y -axis. However, the probability of the normal deviation angles within the standard deviation in three directions is over 90 %, thus the welding surface is recognized as a flat surface.

Similarly, According to Figs. 4(d), (e), and (f), the probability of the normal deviation angles within the standard deviation in three directions is less than 90 %. Among that, more than 30 % of the normal deviation angle is over 10° based on y -axis, thus the

Table 1. Welding process parameters setting.

Parameters	Values
Welding current (A)	200-220
Arc voltage (V)	28-30
Welding speed (cm/min)	15-25
Gas flow (L/min)	25-30
Welding angle ($^\circ$)	45
Diameter of welding wire (mm)	1

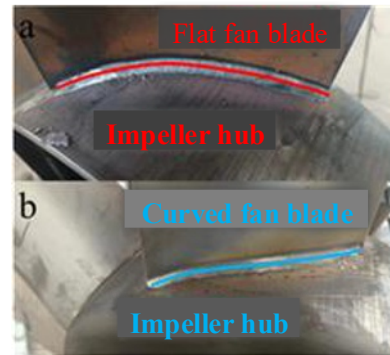


Fig. 7. Actual welding result of impeller: (a) flat fan blade and impeller hub; (b) curved fan blade and impeller hub.

welding surface is recognized as a curved surface. In summary, our algorithm can identify the welding surface with flat and curved surfaces effectively.

4.3 Test and analysis on weld seam recognition

To test the feasibility of weld seam recognition, the recognition experiments on the weld seams of the flat and curved fan blades were performed by our algorithm, respectively, and the results are shown in the red lines in Fig. 5.

To test the recognition accuracy, during the welding path duplication process, the welding robot was stopped randomly to obtain the real-time data, furthermore, its pose was manipulated to obtain the actual position of the weld seam by teaching method, thus the error between the positions of the weld seam recognized by our algorithm and the teaching method was obtained, as shown in Fig. 6. It shows that the errors based on x -axis and y -axis are both less than 1 mm. Particularly, the initial offset of z -axis is set to 2 mm to prevent the welding wire from touching the welded workpieces during the teaching process, thus the error based on z -axis is less than 1 mm. Therefore, the result of recognition on weld seam with our algorithm is almost consistent with the actual position of the weld seam.

To test the actual effect of weld seam recognition furtherly, the welding experiments on the flat and curved blade impellers were performed, respectively, and the welding process parameters were set as Table 1. The welding result shown in Fig. 7 indicates that the weld seam is uniform and flat, without defects such as porosity or cracks.

Furthermore, the weld seam meets the requirements of the welding technology based on the weld seam inspection.

5. Conclusions

In order to improve the accuracy of weld seam recognition of impeller, the corresponding solution and test were performed, and the main conclusions were made as follows:

(1) Based on the shapes of the welded workpieces, a recognition algorithm on weld seam was proposed. According to the rule of the same normal direction for all points in the surface, PCA and normal distribution theory were adopted to recognize the shape of welding surface. Besides, the welding surface was fitted using least square method, and the position of the weld seam was obtained according the intersection line of the two welding surfaces.

(2) A sorting method of welding path points was proposed based on graph algorithm. Firstly, the initial point of the weld seam was determined by graph algorithm. Next the welding path was determined based on bubble sorting method and the distance between the initial point and the weld seam points.

(3) The welding experiments and weld seam recognition of flat and curved fan blades were performed. The results indicate that the position errors of weld seam based on x-axis, y-axis, and z-axis are all within 1 mm, which meets the requirements of welding process.

The proposed automatic recognition method for weld seam has improved the accuracy of weld seam recognition and welding efficiency, and alleviated the labor intensity of workers. However, only a few of workpieces were validated due to the limited conditions. More workpieces need to be studied to further improve the robustness and accuracy of weld seam recognition in the future.

Acknowledgments

This work was supported by Shandong Province Science and Technology Type of Small and Medium-sized Enterprises Innovation Ability Improvement Project (No. 2023TSGC0615), and Tai'an City Science and Technology Development Program, (No. 2024GX053), China.

Conflict of interest

The authors declare that they have no potential conflict of interest.

References

- [1] J. Zeng, G. Cao and Y. Peng, A weld joint type identification method for visual sensor based on image features and SVM, *Nature Reviews Cancer*, 20 (2) (2020) 1-20.
- [2] X. Cao, S. Tuo and G. Zhang, Research on welding seam tracking method and path planning of welding robot, *Gongcheng Kexue Yu Jishu*, 54 (2) (2022) 196-204.
- [3] S. Wang, Y. Zhou and H. Chen, Image processing system of welding seam of steel structure based on laser vision, *Hanji Xuebao*, 43 (2) (2022) 101-105+112+120.
- [4] L. Guo and H. Zhang, Identification tracking welding system for right angle fillet seam in narrow space, *Hanji Xuebao*, 38 (11) (2017) 21-26+130.
- [5] L. Guo and H. Zhang, Discontinuous fold line welding seam recognition and mobile robot tracking system in narrow space, *Jixie Gongcheng Xuebao*, 55 (17) (2019) 8-13.
- [6] X. Li, X. Li and S. Ge, Automatic welding seam tracking and identification, *IEEE Transactions on Industrial Electronics*, 64 (9) (2017) 7261-7271.
- [7] L. Yang, Y. Liu and J. Peng, Advances techniques of the structured light sensing in intelligent welding robots, *International Journal of Advanced Manufacturing Technology*, 110 (1) (2020) 1027-1046.
- [8] W. Hou, D. Zhang and Y. Wei, Review on computer aided weld defect detection from radiography images, *Applied Sciences*, 10 (5) (2020) 1878-1093.
- [9] T. Takubo, E. Miyake and A. Ueno, Welding line detection using point clouds from optimal shooting position, *Journal of Robotics and Mechatronics*, 35 (2) (2023) 492-500.
- [10] F. Wang, C. Liang and X. Han, Visual guidance of welding robot based on weldment recognition and pose estimation, *Kongzhi yu Juece (Control and Decision)*, 35 (8) (2020) 1873-1878.
- [11] Y. Geng, Y. Zhang and X. Tian, A novel welding path planning method based on point cloud for robotic welding of impeller blades, *International Journal of Advanced Manufacturing Technology*, 119 (11) (2022) 8025-8038.
- [12] J. Cao, H. Wu and W. Wang, A visual inspection and classification method for camshaft surface defects based on defect similarity measurement, *IEEE Access*, 12 (2024) 74633-74648.
- [13] L. Yang, Y. Liu and J. Peng, A novel system for off-line 3D seam extraction and path planning based on point cloud segmentation for arc welding robot, *Robotics and Computer-Integrated Manufacturing*, 64 (3) (2020) 1-14.
- [14] J. Liu, W. Li and X. Jia, Operation trajectory extraction method of mobile welding robot based on point cloud reconstruction, *Jisuanji Jicheng Zhizao Xitong*, 64 (3) (2024) 1-14.
- [15] A. Jia, B. Hong and X. Li, Real-time tracking method of 3D zig-zag welding robot swing GMAW based on online trajectory detection, *Jixie Gongcheng Xuebao*, 58 (14) (2022) 116-125.
- [16] Y. Zhang, Y. Jiang and X. Tian, A point cloud-based welding trajectory planning method for plane welds, *International Journal of Advanced Manufacturing Technology*, 12 (125) (2023) 1645-1659.
- [17] J. Yang, C. Wang and W. Luo, Research on point cloud registering method of tunneling roadway based on 3D NDT-ICP algorithm, *Sensors (Basel)*, 21 (13) (2021) 1-18.
- [18] D. Nelson, G. Omar and C. Jhon, Real-time ground filtering algorithm of cloud points acquired using terrestrial laser scanner (TLS), *International Journal of Applied Earth Observation and Geoinformation*, 105 (1) (2021) 102629.
- [19] F. Poux, C. Mattes and Z. Selman, Automatic region-growing

- system for the segmentation of large point clouds, *Automation in Construction*, 138 (1) (2022) 1-22.
- [20] R. Jia and F. You, Application of robust model predictive control using principal component analysis to an industrial thickener, *IEEE Transactions on Control Systems Technology*, 32 (3) (2024) 1090-1097.
- [21] C. M. Shakarji and V. Srinivasan, Theory and algorithms for weighted total least-squares fitting of lines, planes, and parallel planes to support tolerancing standards, *Journal of Computing and Information Science in Engineering*, 13 (3) (2013) 283-302.
- [22] R. Roozegar, N. Balakrishnan and A. Bekker, On multivariate selection scale-mixtures of normal distributions, *Brazilian Journal of Probability and Statistics*, 35 (2) (2021) 351-374.
- [23] L. Guo, Y. He and X. Du, Volumetric measurement of lidar based on visual correction, *Yi Qi Yi Biao Xue Bao*, 44 (10) (2023) 48-59.
- [24] V. Méndez, J. R. Rosell-Polo and M. Pascual, Multi-tree woody structure reconstruction from mobile terrestrial laser scanner point clouds based on a dual neighbourhood connectivity graph algorithm, *Biosystems Engineering*, 148 (1) (2016) 34-47.
- [25] G. Lin, F. Lin and J. Huang, Research on a stereo matching algorithm for substation moving goal oriented to intelligent transportation inspection management, *Dianli Xitong Baohu yu Kongzhi*, 49 (16) (2021) 128-134.



Dongmin Li is an Associate Professor of College of Intelligent Equipment, Shandong University of Science and Technology, Tai'an, China. He received his Ph.D in Mechanical Engineering from Shandong University of Science and Technology. His research interests include pattern recognition, robotics and soft robot,

mechatronics.